

Replacing Phase 1 with Auction-Dynamics MBO

Approach B, measured against the validated deliverables — and a sixth occasion on which an instrument that could not see an effect reported its absence

August 2026

Abstract

Phase 1 (Γ -convergence relaxation by projected gradient descent) costs 22 h–79 h for a completed $N=300$ solution, and its winner-take-all readout is not a valid partition at that scale, so it is repaired afterwards. This report measures **approach B** — volume-constrained multiphase threshold dynamics [18, 9], with the volume constraint imposed by balanced assignment in the manner of auction dynamics [14] — as an outright *replacement* for Phase 1. Four results. (1) At the pinned Phase 2 campaign, B reaches perimeter **184.4118** and **184.1615** at $N=100$ (two seeds) against the control’s 185.2546, and **319.9428** at $N=300$ against 323.3192 — from 214–248 s of Phase 1 against 48,132 s and 79,069 s, a 211–319 $\times$ reduction in the replaced stage. (2) B produces **0 fragmented cells** at both N on its raw labels, where raw PGD at $N=300$ produces 2 and requires the balanced readout; connectivity is the only one of the three validity gates that has content for B, and it is the one B could have failed. (3) The win is attributable to the MBO dynamics and not to the balanced geodesic initialisation B starts from — which is approach C’s first step — by +2.761% against a pre-registered 0.3% threshold. (4) Of thirteen predictions committed before any run, nine held, three missed and one was initially mis-scored. (5) At $N=400$, where no PGD anchor exists *or is possible*, B yields valid, connected, equal-area partitions at $N=400$, 500 and 750 from 920–3,614 s, agreeing with $\sqrt{N}$ scaling on a fixed mesh to better than 0.06% across three points; at $N=750$ the coarsest ladder rung falls *below* the freeze threshold and has to be dropped, which also removes the level-0 fragmentation seen at $N=300$ and $N=500$ — while on the *same* mesh the incumbent needed 206,344 s at $N=300$ and still failed two of the three gates; the absent baseline is the finding, not a gap. The methodological content is again larger than the technical: a **three-part over-merge instrument was built and detected nothing**, and a per-level claim entered the record having never been measured — the sixth instance of this project’s signature artefact.

Provenance

Status	measured
Date	2026-08-19 (scored results); 2026-08-20 ($N=400$, exploratory); 2026-08-22 ($N=400$ re-verification and export)
Surface	torus $T(R=1.0, r=0.6)$ throughout
Branch	feat/phase1-mbo-auction-dynamics
Code	src/partition/mbo_auction.py (the arm); scripts/run_mbo_arm.py (driver); scripts/score_mbo_arm.py (mechanical scorer); testing/test_mbo_auction.py (G1–G8, NC1–NC5b)
Scored runs	results/arm_mbo_20260819_173411_npart100_V114144_seed84172851, results/arm_mbo_20260819_185131_npart100_V114144_seed61803399, results/arm_mbo_20260819_192659_npart300_V47488_seed61803399, results/arm_init_20260819_181203_npart100_V114144_seed84172851
Exploratory runs	results/arm_mbo_20260820_133709_npart400_V114144_seed84172851 results/arm_mbo_20260822_120646_npart500_V114144_seed84172851 and results/arm_mbo_20260822_182903_npart750_V158260_seed84172851 — $N=400$, 500 and 750, no anchor , and none possible (§10). Scored against nothing; reported for gates, cost and absolute perimeter. The $N=500$ and $N=750$ Phase 2 campaigns were run with <code>--profile</code> . $N=750$ uses a <i>different</i> ladder ($162 \times 154 \rightarrow 410 \times 386$, $V=158,260$) and is therefore outside the fixed-mesh $\sqrt{N}$ comparison. $N=500$ predates the arm experiment.yaml snapshot, so it carries none; its config is parameters/torus_500part_mbo.yaml as of commit e8b8b60, which added an mbo: block after the run at the values the run used.
Anchors	$N=100$: run_20260709_081548, Phase 2 best 185.2546144718457 (iterate 20 of 20), raw PGD, Phase 1 48132.12s. $N=300$: run_20260806_123326, Phase 2 best 323.319247087254 (iterate 18 of 19), PGD <i>plus</i> balanced readout, Phase 1 79068.56s. Both from timing_profile.yaml total_wall_s, never run_time_seconds.
Pre-registration	commit 9ffdefb, 2026-08-19 16:53, <i>before</i> the first scored run at 17:34:11. Results commits b414b67, c8aea83, 5bdc947, 54b937c; corrections d4dc66d, d4b75d7.
Figures	make_figures.py from the committed data.yaml, itself re-derivable by extract_data.py.
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1, h5py 3.16.0; macOS 15.3.1 arm64 (Mac mini).
Anchor check	the harness reproduces the $N=100$ control’s own perimeter <i>bitwise</i> (report 07, 0d); no Phase 2 code has changed since.

1 What B is, and what is not claimed for it

Threshold dynamics originates with Merriman, Bence and Osher [18]: alternate short-time diffusion of the phase indicators with a pointwise reassignment, and interfaces move by approximate mean curvature. Esedoğlu and Otto [9] placed the multiphase case on a variational footing with a Lyapunov functional that Γ -converges to a weighted perimeter — the same limit this project’s own energy targets [3, 21, 20]. Jacobs, Merkurjev and Esedoğlu [14] impose *exact volume constraints* by replacing the pointwise threshold with a balanced assignment solved by Bertsekas’ auction algorithm [2]. **None of this is invented here.**

Per mesh level, the arm iterates on a *hard* labelling $\omega : V \rightarrow \{0, \dots, N-1\}$:

1. **diffuse**: solve $(M + \tau K) y_k = M \chi_k$ for each cell, with one prefactorised LU per level and N back-substitutions;

2. **balanced threshold:** $\omega(i) = \arg \max_k [y_{ik} + \psi_k]$, with ψ from the shared semi-discrete optimal-transport dual [1, 17, 15] that approach A already ships.

There is never a continuous field whose winner-take-all readout can diverge from what was constrained. B therefore does not *close* the continuous–discrete gap that motivated approaches A–E; the category is absent.

The regime, and the novelty claim we do *not* make. Corrected 2026-09-17. An earlier version of this paragraph said that every published implementation of this machinery we were aware of convolves on a *uniform Cartesian grid via FFT*, on a flat domain, at small N , and for Pólya’s *outer* problem. That was written from two papers and was wrong on three of its four clauses — two falsified by one paper, the third by another. Reading the lineage itself (docs/math/10-mbo-auction-dynamics/ §2, all copies held at docs/papers/): Jacobs, Merkurjev and Esedoğlu [14, §4.2] already solve the *inner* problem — equal-area minimal-perimeter tessellation — of the *flat 2-torus*, a closed genus-1 surface, at $N = 64$ (recovering the hexagonal lattice) and run the area-preserving flow of a 160-cell Voronoi diagram; Merriman and Ruuth [19, §8.3] evolve five regions of multiphase MBO on an embedded, *curved* torus and 175 regions on the sphere by the closest-point method; and Elsey, Esedoğlu and Smereka [7, 8] run diffusion-generated grain growth at 10^5 grains. Only “uniform grid via FFT” survives, and only for the Euclidean implementations — the graph line [10], whose implicit-Euler step is the one this arm transplants to surface finite elements, never used an FFT, and [14, §5] already run the *constrained* scheme on an unstructured similarity graph. Hu, Liu and Wang [13] remain the closest in *purpose* (auction dynamics as the inner solver of the outer problem), at $n \leq 9$ on $256^2/128^3$ grids, and Bogosel and Oudet [4] use capacity-constrained Voronoi diagrams with > 100 planar cells as an initialisation.

What the held literature does not contain, stated as “we are not aware of” and never as “first”: the scheme on a curved embedded surface discretised by an unstructured triangle mesh with a surface-FEM diffusion step; equal-area constraints at $N = 400$ –1000; an inexact transportation dual in place of the exact auction; and the multilevel mesh ladder. The defensible framing is *combination, surface and scale*, not *method*. The corpus is now the twenty-odd papers listed in docs/papers/README.md, read at the sections cited, plus this project’s bibliography; it is still not a systematic review, and surface-remeshing and computer-graphics venues may cover adjacent ground under different vocabulary.

The difference is load-bearing rather than cosmetic. The FEM operator $(M + \tau K)^{-1}M$ is the *resolvent* of the Laplace–Beltrami operator, not a Gaussian convolution: document 10 derives that one step is mean-curvature flow over time $\tau/2$ rather than τ , that E_τ tends to $\frac{1}{2} \sum_k \text{Per}(\Omega_k)$ rather than $\pi^{-1/2}$ times it, and checks the former on the production mesh; mesh pinning [23] is ours to manage, and no grid-derived τ heuristic is imported.

2 Result 1 — perimeter

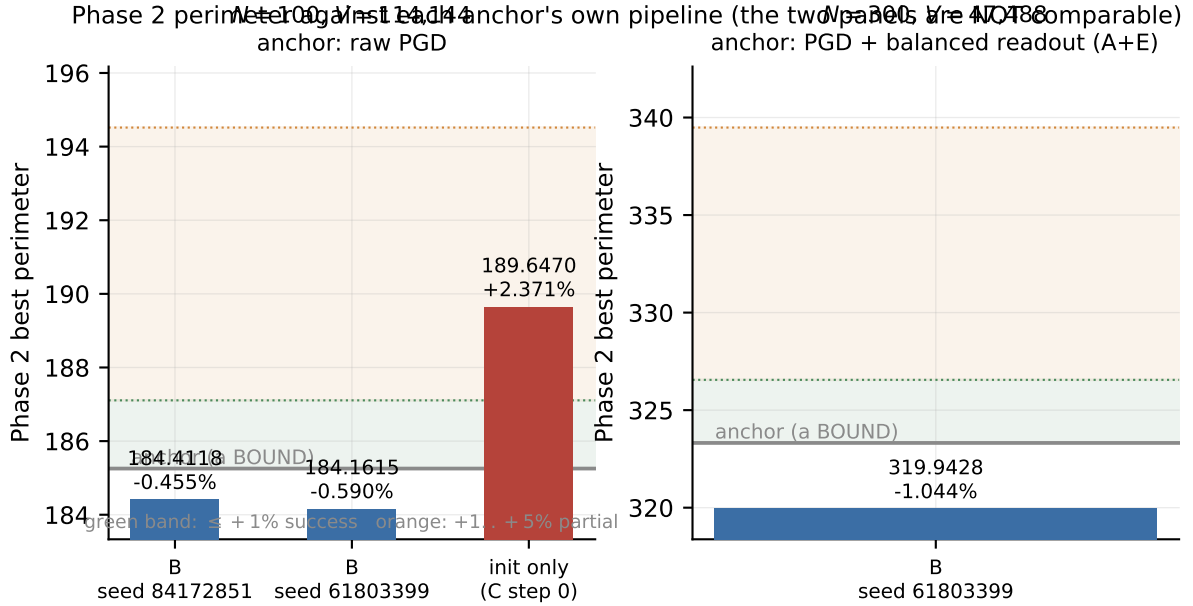


Figure 1: Phase 2 best perimeter against each anchor’s own pipeline. **The two panels are not comparable and must never be pooled:** the $N=100$ anchor is raw PGD, the $N=300$ anchor is PGD *plus* the balanced readout, because raw PGD is not a valid partition at $N=300$.

run	Phase 1	speedup	area worst	frag.	Phase 2 best	vs anchor
$N=100$ s84172851	228 s	210.9×	0.1504%	0	184.4117703525215 (18/20)	− 0.455%
$N=100$ s61803399	214 s	225.2×	0.1493%	0	184.1614647856558 (19/20)	− 0.590%
$N=300$ s61803399	248 s	318.8×	1.4172%	0	319.9427628182375 (16/19)	− 1.044%
init only (C step 0)	67 s	—	0.1355%	0	189.6469917457051 (20/20)	+2.37%

Seed spread at $N=100$ is **0.136%**, which is $3.3\times$ smaller than the margin by which the *worse* of the two seeds beats the control. This is what makes the result scoreable at all: the pre-registration committed that a spread above 1% would mean the primary falsifier could not be evaluated at $\pm 1\%$ and that a single-seed number would be conflating method difference with seed noise.

Six constraints on how these numbers may be stated.

1. **Never pool the two anchors.** They are different baselines. “B beats the best available PGD pipeline at each N , at equal Phase 2 budget” is valid; “B beats PGD by 0.455–1.044%” as a trend in N is not.
2. **The speedup is for the replaced stage only.** End-to-end, including the shared 30–40 min Phase 2, it is $\approx 23\times$ ($N=100$) and $\approx 31\times$ ($N=300$). Against the structure-trigger PGD variant (34,967 s), the incumbent-best figure is $\approx 153\times$.
3. **Neither trajectory is a converged optimum.** Both are at the documented migration-cycling plateau. An earlier draft said “B converged where the control had not”; the sixth review resumed the control’s own Phase 2 for **20 further iterations and found no improvement** (best over all 40 remains 185.2546 at iterate 20; the extension drifts up to ≈ 185.30). B’s plateau is simply 0.455% lower. *That extension is the review’s measurement*

and has not been independently reproduced by us.

4. **PGD’s own seed variance is unmeasured.** Every $N=100$ PGD run across both working directories is seed 84172851, and every $N=300$ $\lambda=11.5$ run is seed 61803399. Each anchor is a single trajectory. “B beats PGD by more than PGD’s own seed lottery” is *not* established.
5. **The $N=300$ margin cannot be decomposed, even in principle.** The readout costs +1.882% in label-boundary length (189.7310 raw $\rightarrow$ 190.7114 shifted $\rightarrow$ 193.3024 repaired, the last being what the anchor’s Phase 2 consumed). B’s raw boundary is 187.2574: shorter than the repaired labels by 3.127% *and* shorter than raw PGD’s own invalid labels by 1.304% — the strong form, since imbalance can only make a boundary artificially short. So the margin is not merely “not paying the repair bill”. But how much is un-erased repair residue cannot be separated, because **no valid repair-free PGD partition exists at $N=300$.**
6. $N=300$ rests on one Phase-2-scored seed.

3 Result 2 — validity, and which gate actually has content

“All three gates pass” is true and misleading

An earlier write-up of this result — ours — said B’s raw labels pass all three validity gates. Phase 0’s own findings say why that is the wrong emphasis:

- **Dormant is vacuous** for any balanced-assignment arm. One-hot labels give peak density 1.0 by construction; the gate cannot fail. The harness already emitted `vacuous_for_arms: True`.
- **Area is near-expected.** Equal area is precisely what B’s assignment step *optimises*. Passing means the solver converged; Phase 0 defined it as the *solver-failure detector*, not a validity test.
- **Connectivity is the only gate genuinely at risk.** Nothing in B enforces it, and the source proposal states it is “not per-step guaranteed”.

The defensible claim is narrower and still strong: **B produces 0 fragmented cells at $N=300$, where raw PGD produces 2** (with 10 imbalanced cells, worst 36.15%) — which is why PGD’s baseline required the balanced readout and B required nothing.

Re-verified through `testing/check_fragmentation.py`, which shares no code with the driver’s reporting: all four arm solutions pass, with **0 cells having more than one raw component including sub-threshold speckle**.

Robustness. Three further $N=300$ MBO ladders at seeds 27182818, 13131313 and 84172851 give **0 fragmented** in every case, with worst area 1.2531/1.4467/1.3887% and label boundaries 186.08/186.55/186.90 — all *shorter* than the scored seed’s 187.2574. Ladder only, no Phase 2; *review-produced, not independently reproduced by us*.

4 Result 3 — attribution: is this B, or is it C’s first step?

B is initialised from one balanced C-scores assignment — farthest-point seeds, multi-source Dijkstra, scores $-d^2$, balanced assignment — which is literally approach C’s step 0 [16, 1, 6]. If MBO barely improved on it, the number in §2 would be a result about C reported under B’s name, and *nothing* in the gate suite would have detected that.

	Phase 1	label boundary	Phase 2 best	censored
init only (C step 0)	67 s	116.1989	189.6469917457051 (20/20)	yes
B (MBO)	228 s	107.6014	184.4117703525215 (18/20)	no
control (PGD)	48,132 s	107.3410	185.2546144718457 (20/20)	plateaued

$$\text{P13} = \frac{189.646992 - 184.411770}{189.646992} = +\mathbf{2.761\%} \quad \text{against a pre-registered threshold of 0.3\%}.$$

Two disclosures. The measured 2.761% is an **upper bound** on MBO’s contribution, because the init arm is censored: its converged value is ≤ 189.6470 , and a lower true value shrinks the gap. For attribution to fail the init would have to reach ≈ 185.0 — another $\approx 2.5\%$ on a trajectory gaining ≈ 0.038 per iterate and decelerating. And the init-only control was run on the *finest* mesh, i.e. the *strongest* member of the init family, which biases the test *against* B.

An unflattering free datum on C. C’s step 0 plus Phase 2 lands at +2.37% above the PGD control — a PARTIAL verdict under this project’s own bands — for 67 s of Phase 1. Real C iterates Lloyd and would improve on this, so it is a *lower bound* on C; but C’s starting point is behind both B and PGD, which is directly relevant to whether C is worth building. Note also that C minimises a *quantisation* energy, not perimeter; the two optima coincide only asymptotically and only in the plane [12, 11], and this torus has Gaussian curvature of both signs.

5 The mesh ladder, and τ

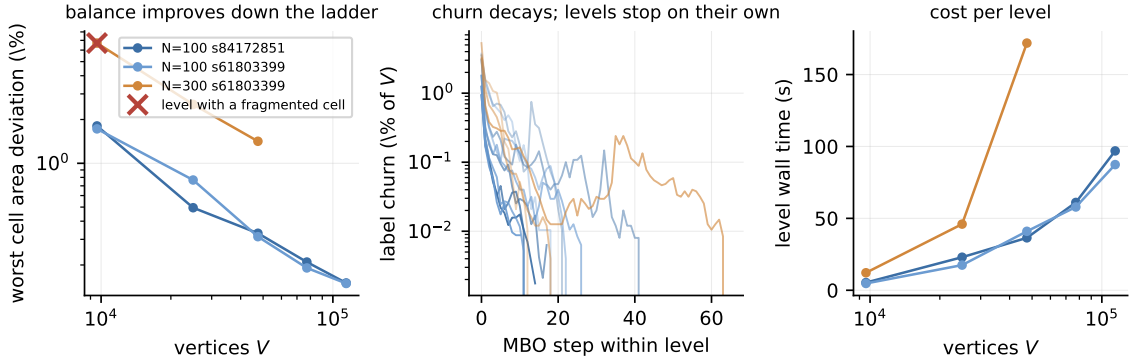


Figure 2: Per-level behaviour. Left: worst cell area deviation improves monotonically down the ladder; the cross marks the one level carrying a fragmented cell. Middle: label churn decays within every level, so levels terminate on their own rather than at the iteration cap. Right: cost per level.

$\tau = \min((ch_{\text{mean}})^2, (\rho R_{\text{cell}})^2)$ with $c = 4$ and $\rho = 1.0$, pinned before any run. $c = 4$ rather than $c = 2$: the latter is the freeze regime [23]. The cap is the over-merge side of a two-sided window, and it binds on exactly one level in use, $N=300$ level 0.

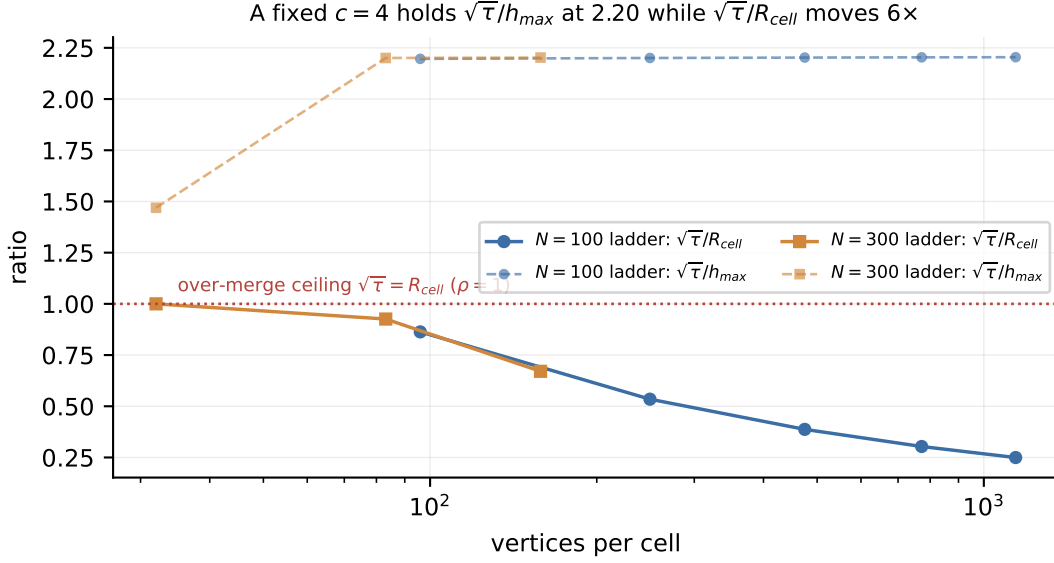


Figure 3: A fixed c does *not* hold the window. $\sqrt{\tau}/h_{\max}$ stays at 2.20 across every level of both ladders while $\sqrt{\tau}/R_{\text{cell}}$ moves $6\times$. Reporting on h_{mean} alone flattens the coarse band: this mesh is $1.81\times$ anisotropic.

6 The headline null: an instrument that detected nothing

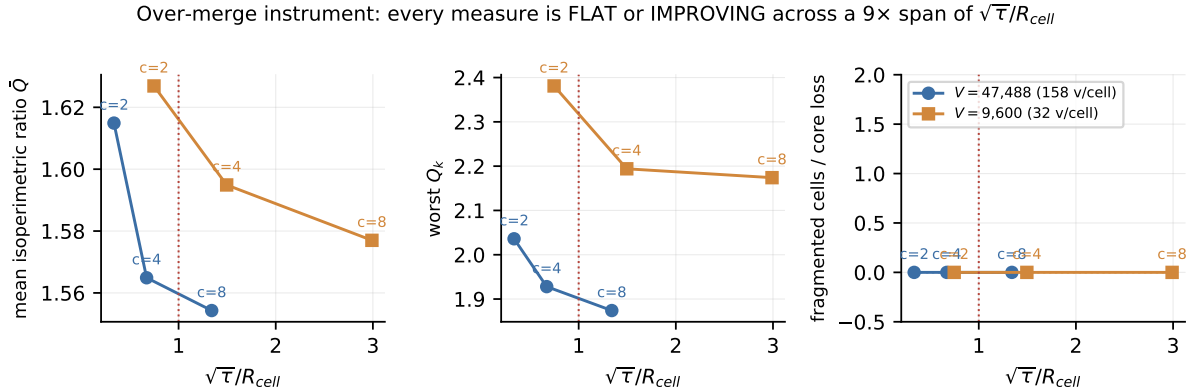


Figure 4: Fragmentation, per-cell isoperimetric ratio $Q_k = P_k^2/(4\pi A_k)$, and core loss, swept over $c \in \{2, 4, 8\}$ at two resolutions. Every measure is flat or *improving*; none registers an over-merge penalty anywhere, including at $\sqrt{\tau}/R_{\text{cell}} = 2.99$, three times the pinned cap.

Assignment quality *cannot* see over-merging — the balanced assignment hits its area target by construction, so a smeared, non-local cell scores as well as a compact one. (An earlier draft of the plan cited a healthy assignment-quality fixture as evidence against over-merging; that was a measurement of a different quantity than the claim it supported, and it was withdrawn.) So three geometry instruments were built. Across a $9\times$ span of $\sqrt{\tau}/R_{\text{cell}}$ and two levels differing $5\times$ in resolution, **not one of them registered anything**: core loss and fragmentation are 0 in all six configurations, and compactness moves monotonically the *other* way.

The instruments are demonstrably *responsive* — $\bar{Q}$ moves 3.8%, Q_{worst} 8%, boundary length 1.9% — they simply move toward improvement. What is established is therefore that **no damage is**

detectable, *not* that over-merging cannot occur.

Standing limit

The ρ question is **open**, and no future claim about c or ρ may lean on this report’s evidence in either direction. The cap was retained because it is pre-registered and because it biases *against* B — it lowers c to 2.68 at $N=300$ level 0, and $c=2$ measures worse than $c=4$. That is a discipline argument, not an empirical one. **Named follow-up:** $c=8$ showed no penalty on any instrument *and* a 0.33% shorter starting boundary — a third of the entire +1% success bar — so the cap may be costing perimeter for no measured benefit. That is a separate arm, scored through the same harness.

7 Why there is no exactly-monotone energy, and what is gated instead

Esedoglu–Otto monotonicity is a minorise–maximise argument requiring (i) the same symmetric positive semi-definite form in the energy and in the assignment objective, and (ii) an *exact* assignment. **Both fail here, for independent reasons.** The theorem’s object is $MA_\tau = M(M + \tau K)^{-1}M$, symmetric PSD; our step maximises $\langle \chi', Dy \rangle$ with $D = \text{diag}(v)$ *lumped*, and DA_τ is not symmetric. And Jacobs, Merkurjev and Esedoglu obtain a theorem because Bertsekas’ auction is exact, whereas `solve_dual_offsets` is subgradient ascent returning its best iterate. **This is the price of a substitution made deliberately in Phase 0** — where Sinkhorn [5] and auction were both measured and both lost to the incumbent on assignment quality — paid here rather than there.

What survives needs neither concavity nor exactness. Since $\omega'(i) = \arg \max_k [y_{ik} + \psi_k]$, multiplying the pointwise optimality by $v_i > 0$ and summing gives

$$\langle \chi', Dy \rangle - \langle \chi, Dy \rangle \geq \sum_k \psi_k (T_k - T'_k)$$

whose right-hand side is exactly the slack that inexactness costs; it vanishes at perfect balance. Measured slack runs 1.7×10^{-4} to 5.5×10^{-3} — five to seven orders of magnitude above the 10^{-9} tolerance an earlier draft proposed for energy monotonicity, so that tolerance was never defensible. **Real E_τ increases do occur** (4 of 96 active steps at $N=300$), and gate G4 therefore gates the inequality, whose measured violation is $\leq 6.13 \times 10^{-8}$, while E_τ is reported as a descriptive trend only.

8 Predictions, and what they caught

	PREDICTED (committed 9ffdefb)	MEASURED	
P1	188.0 (+1.5%), PARTIAL	184.4118 / 184.1615, SUCCESS	MISS
P2	spread < 0.5%	0.136%	HELD
P3	<= 0.28%, point 0.15%	0.1504 / 0.1493%	HELD
P4	<= 3, point 0	0	HELD
P5	>= 30x, point 53x	210.9 / 225.2x	HELD (point missed)
P6	331.4, 80% CI [323.3, 341.0]	319.9428	MISS
P7	<= 2.02%, <= 6 frag, >= 88x	1.4172%, 0, 318.8x	HELD
P8	<= 2 of 5 flagged	0, clauses agree	HELD
P8'	dE flags L2,L3,L4 (falsifying P8)	dE flat 14-22, flags none	MISS
P9	frag not death; worst at L0	1 frag at L0 (295), 0 elsewhere	HIT
P10	viol <= 1e-12; excursions <= ceiling	-6.13e-08; 4/96 within ceiling	HELD
P12	no abort	20/20, 19/19	HELD
P13	>= 0.3%	+2.761%	HELD

Figure 5: Thirteen predictions committed in 9ffdefb at 16:53, before the first scored run at 17:34:11.

P1 and P6 missed, and P1’s *reasoning* is refuted. The rationale was that B sees perimeter only through a coarse- τ surrogate while PGD saw the actual energy for 13.4 h. B also *enters* Phase 2 from a 0.243% *longer* boundary than the control and still finishes ahead — a genuine crossover, so the advantage is not initial geometry. That it is combinatorial is a hypothesis consistent with these data, not established by them. P6 missed worse: its 80% interval’s lower bound was the anchor itself, so essentially no probability was assigned to B beating PGD + A + E at $N=300$.

P8’ missed and invalidated its own model. It predicted, from a log-linear interpolation of $\Delta E/\text{tail}$ in the freeze ratio c/c_{lo} , that levels 2–4 would be flagged pinned. Measured, $\Delta E/\text{tail}$ is *flat* across the ladder — 15.76, 21.62, 15.11, 14.22, 16.49 against a threshold of 79.25 — and level 4, sitting exactly at $c/c_{lo} = 0.999$, scores 16.49 where the model demanded ≈ 260 . The freeze ratio does not predict pinning here.

9 The sixth artefact — and it is ours

Report 07 catalogued five occasions on which a measurement artefact was read as a result about the method. This is the sixth, and it was produced in this work.

The plan asserted “**0 fragmented at every level**” and, on that basis, scored P9 — which predicted over-merge would appear as fragmentation rather than cell death, worst at level 0 — as UNTESTABLE, “the predicted phenomenon never occurred”. **Until the sixth review, LevelReport carried no connectivity field at all.** Per-level fragmentation had never been computed. The claim was an inference presented as a result, and it was false.

With the instrument added and the $N=300$ ladder re-run to a bit-exact reproduction (worst areas 6.7063/2.5394/1.4172%, steps 16/21/66, boundary 187.2574):

level 0: 1 fragmented cell (cell 295). Levels 1 and 2: 0. No cell death.

The phenomenon occurred, at exactly the level P9 named, in exactly the form P9 predicted. **P9 is a hit, not untestable** — “untestable” had absorbed a correct prediction on the strength of a measurement never taken. The stray is healed by level 1’s MBO run, which is direct evidence for the mechanism the taxonomy predicted: stray islands are perimeter-inefficient and shrink under curvature flow, the opposite dynamic to the discrete-area trim that manufactured them.

The fix is instrumentation — `run.gates` per level, seconds of cost, now committed — not a resolution to be careful.

10 Result 5 — $N=400, 500, 750$: existence where the anchor cannot exist

The three results above are all *margins*: B against a control, at a pinned Phase 2 budget. That design has a ceiling built into it. It can only be run where a control exists, and the control is the method whose cost is the reason B was built. Pushed far enough in N , the comparison stops being expensive and starts being impossible.

$N=400, 500$ and 750 are past that point, and this section reports them as what they are: **existence-and-validity** results with no anchor, no margin, and no comparison of perimeters. All three ran in the driver’s exploratory `--config` mode, where the mesh-match assertion is skipped because there is nothing to match. $N=500$ deliberately reuses $N=400$ ’s ladder unchanged, which converts the $\sqrt{N}$ argument below from a single check into a three-point one; $N=750$ could not, for a reason that is itself a result (§10).

The results

$N=400$ and $N=500$ run to $V = 114,144$; $N=750$ runs to $V = 158,260$. All three validity gates pass on the *raw* labels at every N , re-verified with `testing/check.fragmentation.py` independently of `arm.report.yaml`:

	$N=400$	$N=500$	$N=750$
Phase 1 wall	920 s	1,322 s	3,614 s
dormant	0 / 0	0 / 0	0 / 0
area imbalance (bar)	0, 0.7773% (1.1214)	0, 1.0993% (1.4017)	0, 1.2569% (1.5165)
connectivity	0	0	0 at <i>every</i> level
Phase 2 perimeter	368.6603 (it. 19/20)	412.1138 (it. 10/20)	504.9214 (it. 14/20)
variable points	29,287	32,704	47,088
max area violation	8.18e−7	1.10e−6	4.68e−7

All three Phase 2 constraint residuals are converged to tolerance, so the partitions are genuinely equal-area rather than approximately so, and the $N=500$ and $N=750$ bests are *mid-trajectory* — not censored; both descend and then oscillate in the migration-cycling plateau. All three deliverables are exported with `finalised=True`.

The $\sqrt{N}$ cross-check, three points

Not a gate, and the sharpest instrument available in the absence of an anchor. On a *fixed* mesh total perimeter should scale as $\sqrt{N}$; all three runs below sit on the same $V = 114,144$

mesh, so the comparison is free of any refinement effect. $N=500$ was given $N=400$'s ladder unchanged precisely to make this test available, and its predicted value was recorded in `parameters/torus.500part_mbo.yaml` *before* the run.

	predicted	measured	deviation
$N=100 \rightarrow N=400$	368.8236	368.6603	-0.044%
$N=400 \rightarrow N=500$	412.1748	412.1138	-0.015%
$N=100 \rightarrow N=500$	412.3573	412.1138	-0.059%

Three independent runs at three values of N agree with the scaling to better than 0.06%. All three deviations are negative, which is a consistent slight shortfall rather than scatter; with three points we do not attempt to say why.

$N=750$ is *not* in that table, because it runs on a different mesh and the test is only sharp at fixed V . Recorded for completeness and scored against nothing: the fixed-mesh value implied by $N=500$ is 504.7342 and the measurement is 504.9214, +0.037%. Two points worth stating plainly. The magnitude is small, but the *sign* is opposite to all three fixed-mesh checks, and the pre-registered expectation in `parameters/torus.750part_mbo.yaml` was that a finer mesh should land slightly *below* — so the stated direction was wrong. Nothing is concluded from a single unscored point; it is noted so the error is on the record.

The ladder had to change at $N=750$, and the diagnostics said so first

Every earlier config began at 100×96 . At $N=750$ that rung is unusable, and this is visible *before* running: it gives 12.8 verts/cell and $\sqrt{\tau}/h_{\max} = 0.929$ against a local non-freeze threshold of 0.964 — a ratio of **0.964**, **below 1**, so the coarse-triangle band cannot move at all. Every measured run had been above 1 at every level ($N=400$ min 1.049, $N=500$ min 1.067).

The mechanism is a squeeze that tightens with N : more cells means smaller cells, the over-merge cap holds $\sqrt{\tau} \leq R_{\text{cell}}$, so the diffusion length shrinks — while the coarse mesh's triangles do not. Past some N the diffusion no longer reaches across one triangle and the level freezes. **The remedy is the ladder, not τ** : raising τ_c walks toward the over-merge wall on the other side of the same two-sided window. The config therefore keeps levels 1–5 verbatim and drops the dead rung, running $162 \times 154 \rightarrow 410 \times 386$. Every level then sits at ratio ≥ 1.131 , and the finest level's margin (1.131) is better than $N=500$'s (1.109) and $N=400$'s (1.049), both of which ran clean.

It worked, and it bought something beyond avoiding the freeze: $N=750$ is the first high- N run with **zero fragmented cells at every level**, where $N=300$ and $N=500$ each produced one at level 0 that later levels had to heal. Starting at 33 verts/cell instead of 12.8 or 19.2 removes the defect rather than repairing it.

Two mechanisms recur at a third N

Neither is new; both were predicted or observed earlier, and $N=500$ is an independent occasion for them to fail.

Level-0 fragmentation, healed by level 1. $N=500$ level 0 produces one fragmented cell (323) and level 1 has none — exactly the pattern §9 recorded at $N=300$ (cell 295), and the mechanism the taxonomy predicts: stray islands are perimeter-inefficient and shrink under curvature flow. $N=400$ shows none at any level. This is visible only because the per-level instrument now exists; it is the same claim that was once asserted with nothing measuring it.

The over-merge null holds. At $N=500$ the cap binds on two levels and $\sqrt{\tau}/R_{\text{cell}}$ is at its ceiling there, yet the instrument reports core loss = 0 at every level, q_{worst} falling $2.95 \rightarrow 1.81$

down the ladder, and q_{mean} in 1.54–1.61 against $N=400$ ’s 1.56–1.57. A third N , a tighter cap, and the same null of §6. Still undetected rather than excluded.

The mesh-matched companions, and a level 0 below the freeze threshold

The runs above each use the finest ladder their N can support, so $N=750$ and $N=1000$ end at $V=158,260$ and $209,568$ while every earlier deliverable ends at $114,144$. The exported partition carries the Phase 1 mesh, so a downstream study measuring per-cell quantities cannot compare across N when the mesh changes underneath it. Both were therefore re-run on the *common* mesh — and not merely to the same endpoint. Matching only the endpoint would be a different algorithm: in a multi-level scheme the coarse levels do the structural work and the fine levels refine it. The companions use the **identical ladder, seed and τ constants as the $N=400/500$ configs**, changing only `n_partitions`.

That forces 100×96 as the base rung, where the two large N sit *below* the freeze threshold — a regime no previous run had entered:

	level-0 ratio	v/cell	level-0 floor	final worst	bar
$N=750$	0.964	12.8	12.4997%	1.6928%	2.1026%
$N=1000$	0.897	9.6	16.6663%	2.2560%	2.8035%

Both pass all three gates: 0 dead, 0 weak, 0 imbalanced, 0 fragmented. So a sub-threshold level 0 is *survivable* under B. Three observations, the first two of which were not predicted:

(1) **It is a partial, not total, freeze — and it costs real work.** Scoring each level by the fraction of the improvement available to it that it actually captured (its floor being the target it cannot pass), level 0 captures 18% at $N=750$ and 26% at $N=1000$, where every healthy level in both runs captures 55–89%. About a third of normal. This matches the mechanism: the diffusion reaches past the mean edge but not the largest, so fine-triangle regions evolve while the anisotropic coarse band does not.

(2) **The freeze ratio does not predict how much a sub-threshold level accomplishes.** $N=750$ has the *better* ratio (0.964 vs 0.897) and more verts/cell, and was predicted to fare at least as well; it captured *less* (18% vs 26%). With two points no mechanism is offered, but the ratio alone is evidently not the controlling variable below 1.0.

(3) **The damage is manufactured at level 0 and fully healed.** Level 0 produced 2 fragmented cells at $N=750$ and 3 at $N=1000$ — more than any run above threshold — and the chains clear completely: $2 \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow 0$ and $3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 0$. This is the distinction from PGD, where `docs/experiments/06-subfloor-ladder/` measured a starved level 0 leaving a *permanent* runt (46.80% $\rightarrow$ 15.92% across $37\times$ more vertices). Under B the same insult is transient.

Two properties of the common mesh worth recording

Worst-cell error saturates at $\approx 1.61\times$ the granularity floor. On the fixed $V=114,144$ mesh:

	worst	floor	ratio
$N=400$	0.7773%	0.5607%	1.386
$N=500$	1.0993%	0.7009%	1.568
$N=750$	1.6928%	1.0513%	1.610
$N=1000$	2.2560%	1.4017%	1.609

It rises and then plateaus. If that holds, quality on a fixed mesh is predictable from the mesh alone — worst cell $\approx 1.61\times$ granularity, which scales as N/V — so a mesh can be sized for a target area error without running anything. Four points and no mechanism: a pattern, not a law.

A coarser mesh under-measures perimeter. $N=1000$ reads 583.1417 at $V=114,144$ against 583.2558 at 209,568, -0.020% . The direction is not noise: a boundary represented by fewer, longer straight segments is a shorter approximation to the same curve, so the coarse value is an underestimate and the finer one is closer to the truth. Small, but it biases any perimeter-derived quantity systematically with N when the mesh is held fixed — which is the price of holding it fixed.

Equal-area accuracy of the exported geometry is unaffected by any of this. The percentages above are Phase 1’s *discrete* winner-take-all imbalance and do not survive into the export; Phase 2 equalises the actual geometric areas. Measured on the exported best iterates, the worst cell is off target by 0.0014% ($N=400$), 0.0023% (500), 0.0272% (750) and 0.0127% (1000) — the two mesh-matched runs are the loosest of the set, as fewer verts/cell would predict, and all four are negligible.

Why the missing anchor is the finding

The obvious objection is that a result without a baseline is weak. Here it is the reverse, and the reason is measurable rather than rhetorical. Compare at the **matched** finest mesh $V = 114,144$ — the same mesh, so no refinement effect enters:

	PGD, $N=300$ run_20260808_191030	B, $N=400$ arm_mbo_20260820_133709
Phase 1 wall	206,344 s (57.3 h)	920 s
dead / weak	0 / 0	0 / 0
area imbalance	3 imbalanced, worst 24.81%	0, worst 0.78%
fragmented	2	0
valid on raw labels?	no — requires the readout	yes

On the identical mesh, with *one hundred fewer cells*, the incumbent took **224** $\times$ the wall time and failed two of the three gates. Requiring a PGD anchor at $N=400$ therefore asks the incumbent to do precisely the thing whose absence motivated B. The honest claim needs no margin:

Approach B produces a valid, equal-area, fully connected 400-cell partition on a 114,144-vertex torus from 15 minutes of Phase 1, in a regime where the incumbent has already ceased to produce valid partitions at $N=300$ on the same mesh.

What this section does *not* claim

The -0.455% and -1.044% margins of §2 are not extended to $N=400$ or $N=500$, and no perimeter here is compared against PGD. The matched-mesh table above is a statement about *validity and cost*, both measured on both sides at the same V ; it is not a perimeter comparison, and the two must not be blurred. The $\sqrt{N}$ check is B against B, so it tests internal consistency, not correctness against an independent method: three runs agreeing on a scaling law they all obey would not detect an error common to all three. And $N=500$ reuses $N=400$ ’s ladder and seed, so the two are not independent in the mesh or the initialisation, only in the run.

The cost centre has moved — and it is not the solver

Phase 2 took 7,326 s, 8,829 s and **18,323 s** (5.09 h) at $N=400/500/750$, against Phase 1’s 920 s, 1,322 s and 3,614 s. The stage this report replaces is therefore 11%, 13% and 16.5% of end-to-end

cost — the last rising only because $N=750$ was given an extra ladder level. The 211–319 $\times$ of §2 was always a statement about the replaced stage; past $N=400$ the remainder dominates.

The $N=500$ and $N=750$ campaigns were run with `--profile`, and the breakdown is not what the forward plan assumed. **The IPOPT solver accounts for 5.2% of the campaign at $N=500$ and 3.6% at $N=750$ — a *shrinking* share of a growing problem.** At both, IPOPT runs 201 solver iterations per topology iteration, hitting `max_opt_iter:` 200 every time, and the internal callback split is essentially identical (Hessian 40.4% vs 40.7% of solver time). So 95–96% of Phase 2 is spent *outside* `optimize()`: contour rebuild, Steiner setup, migration detection and checkpoint export with its roundtrip verification. **We establish the solver’s share, not the composition of the remainder** — decomposing that is the measurement docs/plans/PUBLICATION_READINESS_PLAN.md Phase 4 should now make, in place of its original premise that the exact-Hessian path was the target.

Phase 2 also scales superlinearly in variable points, close to quadratically. Campaign wall against variable-point count:

	variable points	wall	s/VP	exponent
$N=400$	29,287	7,326 s	0.250	
$N=500$	32,704	8,829 s	0.270	1.69
$N=750$	47,088	18,323 s	0.389	2.00

$N=1000$ has since been measured and extends the table: 62,472 variable points, 29,946 s, 0.479 s/VP, exponent **1.74**. **An earlier version of this paragraph called the trend “superlinear and steepening”; the fourth point refutes the second half** — the exponents run 1.69 $\rightarrow$ 2.00 $\rightarrow$ 1.74, so 2.00 was an excursion, not a trend. The defensible statement is *superlinear, exponent ≈ 1.7 –2.0, with no evident trend*, which is a better outlook than the wording it replaces because it means the cost is not accelerating away.

One cleaner comparison is now available. $N=750$ and $N=1000$ both have mesh-matched companions on $V=114,144$ (§10), giving the first pair measured at *fixed* mesh: 39,704 VPs / 12,908 s and 45,859 VPs / 18,138 s, an exponent of **2.36** — steeper than any of the mixed-mesh figures, which suggests part of what was being attributed to variable-point count was a mesh effect. Two points, so this is a flag for a third measurement rather than a result. Either way Phase 2 is the binding constraint on N .

Measurement note. The $N=750$ mesh-matched campaign was launched twice — the first attempt wrote iterate 1 and stopped — leaving 21 iterate files, two of them bit-identical copies of iteration 1. Its wall time must be taken from the *second* launch (12,908 s); measuring from the campaign directory spans the aborted attempt and reads 14,021 s, 9% high. It is the only campaign of the ten with more than one launch.

A reporting trap in `timing_profile.yaml`

The Phase 2 profile’s `summary.total_wall_s` is 459.27 s while the campaign took 8,829 s. The field accumulates only time inside `optimizer.optimize()`; everything else in the loop is outside it. A **19 $\times$** under-report if read as campaign wall — the same class of trap as `run_time.seconds` in the Phase 1 metadata, which under-reports by 3.6 $\times$ and is already documented. Campaign wall must be taken from the iterate timestamps.

Two scaling consequences worth recording. The τ over-merge cap binds on *two* levels at both N (against one at $N=300$ and none at $N=100$) as cells shrink relative to the coarse mesh. And the vertex-granularity floor has risen from 0.5607% to 0.7009%, pushing the worst cell from 1.39 $\times$ to 1.57 $\times$ that floor against a 2 $\times$ bar — so the finest mesh will have to grow before the bar is reached, and that raises Phase 2, which is now the expensive half.

11 Threats to validity that no instrument here addresses

1. **PGD’s seed variance** (§2, constraint 4). Both anchors are single trajectories; the corpus was enumerated across *both* working directories, since `results/` is gitignored and each worktree has its own.
2. **Over-merging** (§6). Not excluded, merely undetected.
3. **Phase 2 budget as a confound**. B may start from geometry that lets Phase 2 converge *within* a budget PGD cannot use as well. The 40-iteration control extension argues against this, but is review-produced.
4. **Same-machine identity** of the July control and the August arms is consistent with the recorded hardware but is not stamped in run metadata.
5. $N=1000$ is untouched. Nothing here establishes that B’s behaviour extrapolates.

12 Conclusions

1. B replaces Phase 1 at $211\text{--}319\times$ less wall time in the replaced stage and reaches a *shorter* Phase 2 perimeter than the best available PGD pipeline at both N , at equal Phase 2 budget.
2. Its raw labels are connected at both N — the one validity gate B could have failed — so the readout and repair stages are unnecessary for B.
3. The advantage is attributable to the MBO dynamics, not to the balanced geodesic initialisation, by $+2.761\%$ against a 0.3% threshold.
4. Three pre-registered predictions missed, all in B’s favour, and one was initially mis-scored in B’s *disfavour* by an instrument that did not exist.

Provenance of the references

Updated 2026-09-17. Every paper cited in §1 is now held at `docs/papers/` and was read at the sections cited, not only at the abstract; the reading is recorded, with a table of cell counts, domains and constraints, in `docs/math/10-mbo-auction-dynamics/` §2. The original note stands for the two papers it covered: [13] is Shilong Hu, Hao Liu and Dong Wang (v1, 25 May 2024) — an earlier draft of this report guessed two of the three given names wrongly, from a secondary description — and [4] is Benjamin Bogosel and Edouard Oudet (v2, 7 June 2021). The consequence of the wider reading is the corrected novelty paragraph in §1.

Of the entries in `docs/math/shared/references.bib` that this report cites, every one is now verified against a held copy ([22], the last second-hand entry in the lineage, was obtained and verified on 2026-09-22). Entries this report does not cite keep the upstream note that volume and page numbers should be checked before external distribution.

Related documents

- Plan, pre-registration and full correction history: `docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md`
- Taxonomy: `docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md`
- The instrument this report is scored through: `docs/experiments/07-phase0-shared-harness/`

- The gap B makes vacuous: docs/reference/winner_take_all_partition_gap.md

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